Reflux in Special Groups: Neurological Impairment, Oesophageal Atresia, Cystic Fibrosis and Obesity.

In the last few chapters we built a careful approach to gastro-oesophageal reflux in children. We separated physiological reflux from disease, we learned not to treat a crying infant with acid suppression, we looked at how to diagnose reflux disease in older children, and we discussed how to stop proton pump inhibitors once they are no longer needed. All of that was built on one guideline, the 2018 paediatric reflux guideline of the North American and European societies for paediatric gastroenterology, hepatology and nutrition, known as NASPGHAN and ESPGHAN. There is something important written near the beginning of that document. It says that it was not designed for children with neurological impairment, with repaired oesophageal atresia or with cystic fibrosis, and it points the reader to separate guidance for them. This chapter is about those children, and about one more large group, children with obesity.

The clinical question is this. When a child with cerebral palsy, a child whose oesophagus was repaired at birth, a child with cystic fibrosis, or a teenager with obesity has reflux, do the usual rules apply? The main idea of the chapter is that they do not. Each group has its own mechanism of reflux, its own complication that we are trying to prevent, and its own threshold for testing and for surgery. In neurological impairment, reflux should be proved objectively before long-term treatment or surgery, and a fundoplication should not be added reflexively when a feeding gastrostomy is placed. After oesophageal atresia, reflux is lifelong and needs endoscopy on a schedule, but prophylactic acid suppression does not prevent strictures of the anastomosis. In cystic fibrosis, reflux is common and often silent, and acid suppression may carry a respiratory cost. In obesity, the weight is the main treatment, and the choice of bariatric operation matters.

A few terms first. A proton pump inhibitor, usually shortened to PPI, is a drug such as omeprazole or esomeprazole that blocks acid secretion in the stomach. It reduces the acidity of refluxed material, but it does not reduce the number of reflux episodes. Multichannel intraluminal impedance with pH monitoring, usually called pH-impedance, is a twenty-four hour test with a thin catheter in the oesophagus that detects every reflux episode, acid or not, and relates it to symptoms. A fundoplication is an operation that wraps the top of the stomach around the lower oesophagus to create a tighter barrier. A gastrojejunal tube is a feeding tube placed through a gastrostomy and passed beyond the stomach into the jejunum, so that feeds bypass the stomach altogether.

The sources for this chapter include the 2017 ESPGHAN guideline on gastrointestinal and nutritional complications in children with neurological impairment, written by Romano and colleagues; the 2018 NASPGHAN and ESPGHAN reflux guideline by Rosen and colleagues;

the 2016 joint ESPGHAN and NASPGHAN guideline on gastrointestinal complications of oesophageal atresia by Krishnan and colleagues; the 2020 consensus of the European Reference Network for rare inherited congenital anomalies, known as ERNICA; the classic surgical series of Pearl and colleagues from 1990; large American cohort studies by Srivastava and colleagues in 2009 and Barnhart and colleagues in 2013; a meta-analysis by Lauriti and colleagues in 2018; meta-analyses on strictures by Miyake and colleagues in 2018 and Wyllie and colleagues in 2023; a Dutch study of adults after atresia repair by Vergouwe and colleagues in 2018; cohort studies of acid suppression in cystic fibrosis; studies of obesity and reflux from Italy, California and Hungary; the Teen-LABS study of adolescent bariatric surgery; a study of pantoprazole dosing in obese children by Shakhnovich and colleagues; and the adult guidance of the American College of Gastroenterology from 2022 and the American Gastroenterological Association from the same year.

Part one. Why the barrier fails differently.

The antireflux barrier is the lower oesophageal sphincter, helped by the crura of the diaphragm and the angle at which the oesophagus enters the stomach. In every child, most reflux happens during transient relaxations of the sphincter. What changes in these four groups is the load placed on the barrier.

In a child with neurological impairment, many things push in the same direction. Hypotonia and poor coordination slow the movement of the gut. Gastric emptying is often delayed, so the stomach stays full for longer. Scoliosis distorts the anatomy around the diaphragm. These children often spend most of the day lying down, which removes the help of gravity. Constipation and spasticity raise the pressure inside the abdomen. And because swallowing is often impaired, both refluxed material and saliva can be aspirated into the lungs.

After oesophageal atresia, the oesophagus itself is abnormal. It is intrinsically dysmotile, so it clears refluxed acid slowly. The repair can shorten the segment of oesophagus that sits inside the abdomen or place it under tension, which weakens the barrier. And the anastomosis, the join made by the surgeon, is prone to narrowing, so reflux may cause or worsen a stricture.

In cystic fibrosis, the lungs drive the reflux. Chronic cough and the strongly negative pressure in the chest during breathing amplify the transient sphincter relaxations. Gastric emptying can be delayed, the head-down positions used in chest physiotherapy promote reflux, and bile acids from the duodenum can reach the airway. In obesity, raised pressure inside the abdomen, more transient relaxations and a higher rate of hiatal hernia all contribute, and central fat around the waist matters more than body weight alone.

This matters because the mechanism decides the goal. In neurological impairment and cystic fibrosis, the organ we are trying to protect is the lung. After atresia, it is the oesophagus itself, for life. In obesity, it is the weight. And because a PPI reduces acid but not the number of

reflux events, it cannot on its own fix a problem driven by volume, posture or dysmotility.

Part two. Children with neurological impairment.

The 2017 ESPGHAN guideline states that reflux disease has been reported in as many as seventy percent of children with neurological impairment. The difficulty is that these children often cannot describe heartburn. Their symptoms are behavioural: arching, grimacing, disturbed sleep, refusal of food and recurrent chest infections. The same behaviours are caused by constipation, dental pain, fractures and dislocated hips, so a careful search for other sources of pain should come before we attribute them to reflux.

The guideline makes several recommendations. It advises that reflux disease should be diagnosed with objective tests, meaning pH or pH-impedance monitoring and, or, upper endoscopy. Because reflux is so common and these children are fragile, it accepts a trial of a PPI with close follow-up as an alternative, but the point is that long-term treatment or surgery should not rest on guesswork. PPIs are recommended as first-line treatment, a strong recommendation, and long-term treatment should be re-evaluated from time to time. Prokinetic drugs are not recommended for routine use and may be considered only when reflux is uncontrolled. Baclofen, which reduces transient sphincter relaxations, has been studied in only eight children, in whom it reduced vomiting but did not change acid exposure. A trial of a whey-based formula is suggested for children with reflux, gagging and retching, and thickening can be added to other treatment.

The big question in these children is surgery. For decades many surgeons added a fundoplication whenever they placed a feeding gastrostomy. The evidence now points the other way. Start with what a fundoplication achieves in this group. In the series published by Pearl and colleagues in 1990, one hundred and fifty-three neurologically impaired children were compared with eighty-one neurologically normal children after antireflux surgery. Late complications occurred in twenty-six percent of impaired children against twelve percent of normal children. Reoperation was needed in nineteen percent against five percent. The overall failure rate was twenty-eight percent against six percent. And deaths related to aspiration occurred in nine percent against one percent. Herniation of the wrap accounted for more than half of the reoperations. A meta-analysis by Lauriti and colleagues in 2018 found the same direction with modern data: recurrent or persistent reflux in about fourteen percent of impaired children against about nine percent of normal children, and redo fundoplication in about thirteen percent against nine percent. The ESPGHAN guideline cites a reoperation rate of about fifteen percent in impaired children against seven percent in others. In short, a fundoplication in these children fails two to three times as often.

What does the wrap change? Srivastava and colleagues, writing in the British Medical Journal in 2009, compared admissions before and after fundoplication in neurologically impaired children in forty-two American hospitals. After surgery, the rate of admissions related to reflux

fell by about thirty percent, aspiration pneumonia admissions fell by about thirty percent, and admissions needing mechanical ventilation fell by about sixty percent. But admissions for pneumonia of any kind did not change, with a rate ratio of about one point zero seven, and admissions for asthma actually rose by about half. This was a before-and-after comparison without a control group, but the pattern makes sense. A wrap reduces some reflux-related events, but it cannot stop a child from aspirating his own saliva.

Is it worth adding a fundoplication at the time of gastrostomy? Barnhart and colleagues, writing in JAMA Pediatrics in 2013, studied four thousand one hundred and sixty-three neurologically impaired infants who had a gastrostomy placed in a neonatal intensive care unit. About a third also had a fundoplication. Strikingly, fewer than one in ten of those who had a fundoplication had a pH study beforehand, and only about one in twenty-five had an endoscopy. In the first year, admissions related to reflux occurred at a rate of about zero point nine two per child after gastrostomy alone and one point zero two after gastrostomy with fundoplication, so there was no benefit. Admissions for aspiration pneumonia were the same in both groups. An Italian series of one hundred and forty-five children, published in 2021, found that only six percent needed a fundoplication later after gastrostomy alone, and other series report five to ten percent. A Cochrane review in 2013 found no randomised trials of fundoplication at the time of gastrostomy. On this basis, the ESPGHAN guideline makes a strong recommendation against routine antireflux surgery when a gastrostomy is placed. The practical sequence is to place the gastrostomy, treat reflux medically, and reassess with objective testing if symptoms persist.

When medical treatment truly fails, the choice lies between a fundoplication, feeding through a gastrojejunal tube, and, rarely, total oesophagogastric dissociation, an operation that disconnects the oesophagus from the stomach and drains it into a loop of bowel. The Lauriti meta-analysis compared these options. Between fundoplication and gastrojejunal feeding, recurrent reflux was similar, about thirty-five percent against thirty-one percent, and pneumonia was similar, about seventeen percent against twenty-two percent, but complications were more common with fundoplication, about fifty-two percent against thirty-three percent. An American cohort published in Pediatrics in 2009, comparing three hundred and twenty-three children after fundoplication with forty-three children on gastrojejunal tubes, concluded that neither option was clearly superior. Total oesophagogastric dissociation almost abolished recurrent vomiting compared with fundoplication, about one percent against twenty-five percent, but it is a major operation, and the guideline reserves it for very selected children. Gastrojejunal tubes have their own burden: they migrate, and the ESPGHAN guideline quotes a mean functional duration of only about thirty-nine days, so they need frequent replacement in radiology. Both guidelines accept jejunal feeding as an alternative to fundoplication. The right choice is often the one the family can live with. If a fundoplication is planned, the guideline advises an endoscopy with biopsies first, and benefit is greatest when

there are respiratory complications.

Part three. Children after oesophageal atresia.

Reflux follows between a quarter and four fifths of oesophageal atresia repairs, depending on the series and the method of detection. The reasons were described above: an intrinsically dysmotile oesophagus with poor clearance, and a repair that may shorten or tense the segment inside the abdomen. A second key fact is that symptoms correlate poorly with mucosal injury in these children. For that reason, the 2016 joint guideline relies on endoscopy rather than symptoms.

The guideline recommends a PPI from the neonatal period up to the first year of life, or longer if reflux persists, although the evidence for this is graded as low. It states that endoscopy with biopsies is mandatory for monitoring reflux, which is graded as high evidence. It suggests a schedule of endoscopy after stopping the PPI, before the age of ten and at transition to adult care. It finds pH-impedance useful in selected children. It supports fundoplication for recurrent strictures or for reflux poorly controlled on maximal PPI therapy. It does not recommend routine screening for strictures, or routine dilatation, in children without symptoms. And it notes that Barrett oesophagus, a change in the lining that predisposes to cancer, is more frequent in adults who were born with atresia.

The most debated recommendation is the prophylactic PPI for the first year. The idea is that acid reflux onto a fresh anastomosis promotes scarring and stricture. The data do not support it. A meta-analysis by Miyake and colleagues in 2018 found strictures in about forty-five percent of infants given prophylactic PPI and forty-four percent of those not given it, an odds ratio of about one point one, with very low-quality evidence. A larger meta-analysis by Wyllie and colleagues in 2023, pooling twelve observational studies and one thousand three hundred and ninety-five infants, found an odds ratio of about one point three, with a confidence interval from about zero point nine to about one point nine, so again no protective effect. A prospective American study from the Midwest Pediatric Surgery Consortium in 2022 followed one hundred and fifty-six infants. Ninety percent received acid suppression, and still about half developed strictures, with no relation to the duration of acid suppression. The ERNICA consensus of 2020 reflects this uncertainty. Its experts agreed on routine acid suppression after repair, on a PPI as the agent and on tapering at the end, but a fixed course of twelve months reached only about sixty-seven percent agreement, below the seventy-five percent threshold needed for consensus. They also agreed on a pH or pH-impedance study when acid suppression is stopped. A multicentre randomised trial in the United Kingdom, called TOAST, is now comparing acid suppression with none for preventing strictures, and received ethics approval in 2025. Until it reports, the one-year prophylactic course is best understood as a habit awaiting a trial, rather than proven prevention.

Reflux after atresia does not end at school age. In one study, about one child in five had histological abnormalities after the PPI was stopped. In the Dutch study by Vergouwe and colleagues, one hundred and fifty-one adults who had been born with atresia underwent endoscopy. Barrett oesophagus was found in about six and a half percent, roughly four times the rate in the general population, and squamous cell carcinoma in about zero point seven percent, about a hundred times the expected rate. Eosinophilic oesophagitis is also more common in these children, reported in between about one and seventeen percent of series, and it is managed as in other children. The ERNICA consensus therefore recommends endoscopy at one year, at least two more endoscopies before transition, and then, in adults, endoscopy with biopsies from four quadrants at transition and every five to ten years afterwards. In practice, every child with repaired atresia needs a written surveillance plan that travels with them to adult care. And dysphagia in an adolescent with repaired atresia is a stricture, an impacted food bolus or eosinophilic oesophagitis until proved otherwise.

Part four. Children with cystic fibrosis.

Reflux is common in cystic fibrosis. Reviews quote a range of about thirty-five to eighty-one percent of patients, depending on the method. In individual pH-impedance studies of children, pathological reflux was found in about two thirds in one series, about fifty-five percent of forty-four children in a Polish study published in 2015, about forty-six percent in another series, and about a third of twenty-nine children under ten in a Brazilian study published in 2023. The most important finding is that it is often silent. In the Polish study, only fourteen of the forty-four children had typical symptoms.

Why does this matter? Because the damage is done in the lungs. In the Brazilian study, slower clearance of acid from the oesophagus correlated with pulmonary exacerbations. Aspiration of bile acids has been linked to higher levels of neutrophil elastase in the airway and to lower lung function. In patients after lung transplantation, fundoplication within ninety days of transplant has been associated with better survival of the graft. The 2018 reflux guideline names cystic fibrosis as a high-risk group in which pH testing on treatment may help, and it suggests wireless pH monitoring for children with cystic fibrosis and chronic cough.

The paradox is acid suppression. PPIs are prescribed very widely in cystic fibrosis, both for reflux and to help pancreatic enzymes work, because enzymes are less active in an acidic duodenum. But the evidence that PPIs help the lungs is absent, and there are signals of harm. A small randomised trial published in 2014 gave esomeprazole or placebo to seventeen adults. There were five exacerbations in nine patients on esomeprazole and two in eight on placebo, a difference that was not significant, and there was no benefit in lung function. A cohort of one hundred and fourteen patients published in 2017 found that PPI use was independently associated with more hospital admissions. And a Dutch cohort of five hundred and forty-five children, published in the European Respiratory Journal in 2018, found that PPI use was

associated with a faster decline in forced expiratory volume in one second, a standard measure of lung function, and with more exacerbations later on. Confounding is likely, because sicker children are more likely to be given PPIs, but no trial shows benefit. The newer drugs that correct the cystic fibrosis protein, called modulators, have not yet changed this picture: in a study of one hundred and twenty-four children aged six to eleven, gastrointestinal symptom scores did not change significantly over twelve months of modulator treatment. There is no guideline on reflux from the Cystic Fibrosis Foundation or the European Cystic Fibrosis Society. A sensible approach is to use a PPI for proven acid disease or for malabsorption that persists despite adequate enzyme dosing, to review it at every visit, and to stop it when the indication ends.

Part five. Children with obesity.

The fourth group is the largest. Several studies show that reflux symptoms rise with weight, although not consistently. In an Italian study published in 2012, reflux symptoms were reported by about sixty-one percent of obese children against about twenty-two percent of children of normal weight, and by about sixty-two percent of children with a waist circumference above the ninetieth centile against about twenty-four percent of those below the seventy-fifth. The questionnaire used was not validated. A Californian study of six hundred and ninety thousand three hundred and twenty-one children aged two to nineteen, published in 2011, found that a diagnosis of reflux was about thirty percent more likely with moderate obesity and about forty percent more likely with extreme obesity, compared with normal weight, but there was no association in children aged two to five. Other studies are negative. An American endoscopic series in 2010 found oesophagitis in about twenty-four percent of overweight children and about twenty-five percent of normal-weight children, and a Hungarian questionnaire study in 2024 found reflux in about twenty-nine percent of obese and twenty-seven percent of non-obese children. Taken together, symptoms rise with weight, central fat around the waist seems to matter more than the body mass index, mucosal injury is less clearly related, and the association appears after early childhood.

What should we do? In adults, the 2022 guideline of the American College of Gastroenterology makes a strong recommendation, based on moderate evidence, for weight loss in overweight or obese patients with reflux. It also makes conditional recommendations to raise the head of the bed and to avoid meals in the two to three hours before bedtime. We have no paediatric trials showing that weight loss improves reflux, but the adult evidence and the mechanism are both strong.

Two further points matter for paediatricians. The first is dosing. Shakhnovich and colleagues showed in 2018 that dosing pantoprazole by total body weight gives higher drug exposure in obese children, and that dosing by lean body weight normalises exposure in children aged twelve and older, and only partly in younger children. So a PPI in an obese child should be

dosed by lean or ideal weight, and capped at the adult dose. The second is surgery. In the Teen-LABS study of adolescents who had bariatric surgery, reflux symptoms rose from about nine percent before surgery to about twenty-seven percent eight years after a sleeve gastrectomy, while after gastric bypass they barely changed, from about twelve percent to thirteen percent. After adjustment, a sleeve was associated with about two and two thirds times the odds of reflux symptoms compared with bypass. This agrees with the 2022 clinical practice update of the American Gastroenterological Association, which says that Roux-en-Y gastric bypass is an effective antireflux operation in obesity and that sleeve gastrectomy has the potential to worsen reflux. For an adolescent with reflux who is being considered for bariatric surgery, that difference should enter the decision.

Part six. Different goals, and what the guidelines say.

It helps to put the four groups side by side. In neurological impairment, the goal is to prevent aspiration and to keep the child comfortable. Test with pH-impedance or endoscopy before committing to long-term therapy, start with a PPI, a whey-based formula and treatment of constipation, and reserve surgery for failed medical therapy and respiratory complications, not for the moment a gastrostomy is placed. After oesophageal atresia, the goal is to protect the oesophagus for life. Test with endoscopy on a schedule and pH-impedance when the PPI is stopped, use a PPI in the first year and afterwards according to findings, and consider surgery for recurrent strictures or reflux uncontrolled on maximal treatment. In cystic fibrosis, the goal is to protect the lungs. Test with pH-impedance when there is cough or declining lung function, use the shortest effective course of PPI, and consider surgery around lung transplantation. In obesity, the goal is to treat the cause. Endoscopy is needed only for alarm features, the first-line treatment is weight loss with a PPI dosed by lean weight, and if bariatric surgery is planned, bypass is preferable to sleeve. The healthy-child approach of an empirical PPI trial followed by watchful waiting fits none of these groups without adjustment.

How do the guidelines compare? On objective testing, the documents are aligned: the 2018 guideline suggests pH monitoring on treatment in high-risk children, the neurological impairment guideline asks for pH-impedance or endoscopy or a closely watched PPI trial, and the American adult societies advise objective testing first when symptoms are only outside the oesophagus. On fundoplication at the time of gastrostomy, they are also aligned: the 2018 guideline accepts jejunal feeding as an alternative to surgery, and the neurological impairment guideline recommends strongly against routine surgery. On prophylactic PPI after atresia, there is a material conflict: the 2016 guideline recommends a course to one year, ERNICA could not agree on its length, and three pooled analyses show no fall in strictures. On surveillance after atresia, the documents differ: the paediatric guideline and ERNICA set out specific schedules, while adult rules for Barrett oesophagus were not written with these patients in mind. On reflux in cystic fibrosis, there is simply a gap, since no cystic fibrosis society has written guidance. And on weight loss in obesity, adult guidance is strong while the paediatric guideline

says nothing specific. An update of the 2016 atresia guideline has been approved, but it had not been published at the time of writing.

Part seven. Traps and pearls.

Several errors recur. The first is to add a wrap whenever a gastrostomy is placed: admissions are not reduced, and most children never need one. The second is to believe that a fundoplication stops aspiration: saliva is still aspirated, and overall pneumonia admissions did not change. The third is to believe that a PPI prevents strictures after atresia repair: the meta-analyses are null, and strictures should be dilated when they cause symptoms. The fourth is to discharge a well adolescent after atresia from follow-up: Barrett oesophagus is found in about one adult in fifteen, and surveillance is for life. The fifth is to treat a PPI as harmless in cystic fibrosis: it has been linked to faster decline in lung function, and it should be reviewed at every visit. And one more: dosing a PPI by total body weight in an obese adolescent overexposes the child, so use lean weight.

Let us apply this to some common situations. A child with cerebral palsy is about to have a gastrostomy, and the surgeon offers to add a wrap. The best answer is to place the gastrostomy alone and to treat reflux medically, because a wrap did not reduce admissions. An impaired child who arches and sleeps badly should first be checked for constipation, pain and fractures, and only then tested with pH-impedance, because the symptoms are not specific. A child with recurrent aspiration despite a PPI is the right candidate for a discussion of jejunal feeding or fundoplication, since pneumonia rates are similar and the burdens differ. An infant who has completed twelve months of PPI after atresia repair should have it tapered and stopped, with pH-impedance and endoscopy, as ERNICA advises. A teenager with repaired atresia who is moving to adult care needs an endoscopy with biopsies and a written plan. A child with cystic fibrosis on a long-term PPI without a clear reason should have a trial off it, with testing if symptoms return. And an obese fifteen-year-old with heartburn needs a weight programme, with a PPI dosed by lean weight if one is needed.

So, tomorrow in clinic, carry these points. The 2018 reflux guideline excludes children with neurological impairment, oesophageal atresia and cystic fibrosis, so use the group-specific documents. Reflux affects up to seventy percent of neurologically impaired children, and it should be proved before long-term therapy or surgery. After fundoplication in these children, reoperation was needed in nineteen percent against five percent of normal children, and aspiration deaths occurred in nine percent against one percent. Adding a fundoplication at gastrostomy did not reduce reflux admissions, and only about one child in ten needs one later. After a wrap, aspiration pneumonia admissions fell, but pneumonia of all kinds did not. After atresia repair, reflux affects between a quarter and four fifths of children, and endoscopy, not symptoms, guides care. Prophylactic PPI does not prevent strictures. Adults born with atresia have Barrett oesophagus in about six and a half percent and squamous carcinoma in about zero

point seven percent, so endoscopy continues every five to ten years. In cystic fibrosis, reflux is common and often silent, and PPI use has been linked to declining lung function and more exacerbations. And in obesity, weight loss is the treatment, a sleeve gastrectomy triples reflux symptoms over eight years, and a PPI should be dosed by lean weight. Reflux is one word, but in these four groups it names four different problems, and each deserves its own plan.